

E-commerce Sentiment Analysis

Automating Feedback Intelligence with NLP

The Challenge

Manual review of thousands of comments is slow, expensive, and inconsistent. We need speed to save customer satisfaction.

**BEST CUSTOMER
REVIEW PLATFORMS
FOR ECOMMERCE BUSINESSES
2026**

Automation Pipeline

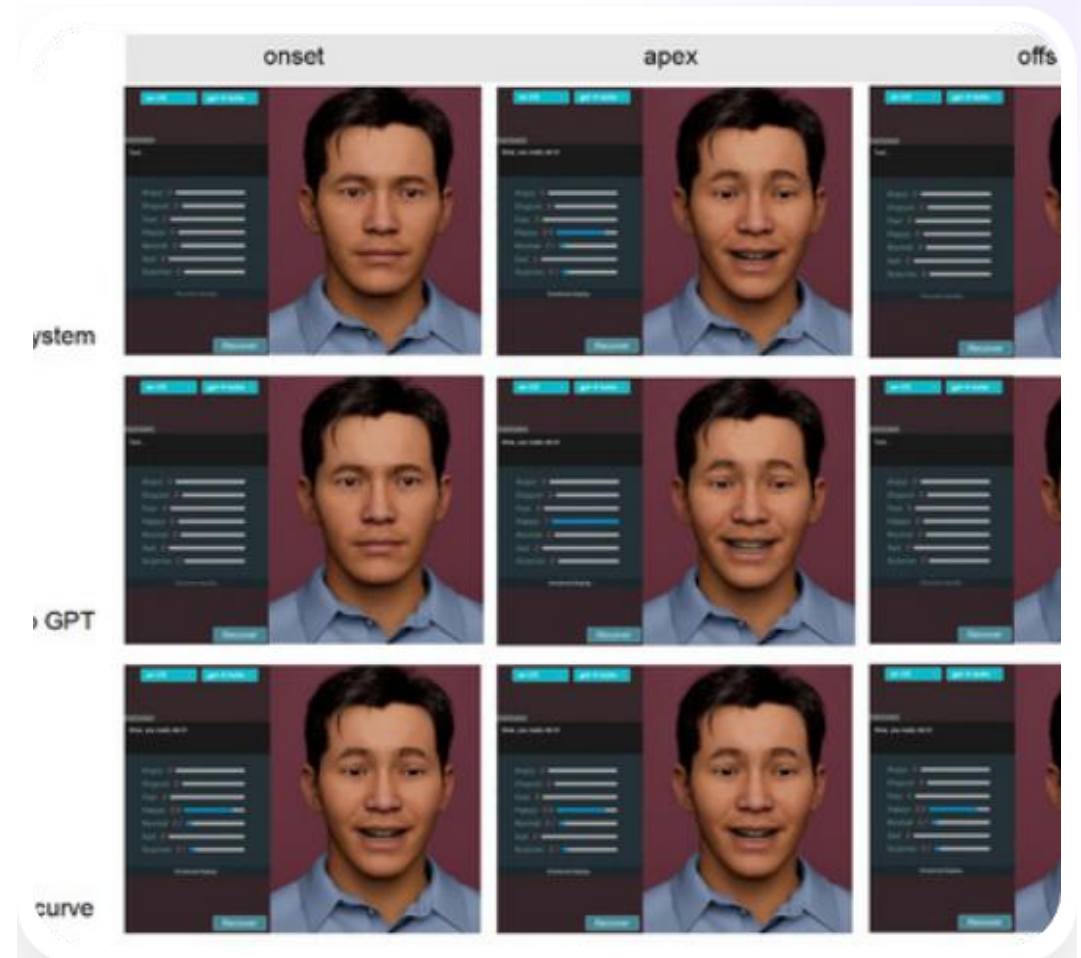
Using **Python** and **Natural Language Processing (NLP)**, we built a pipeline to ingest raw e-commerce reviews and output actionable data.

1. Data Ingestion

Reading massive .xlsx datasets with 8,000+ entries.

2. Sentiment Mapping

Algorithmic segregation into Positive & Negative labels.



Analysis Scale

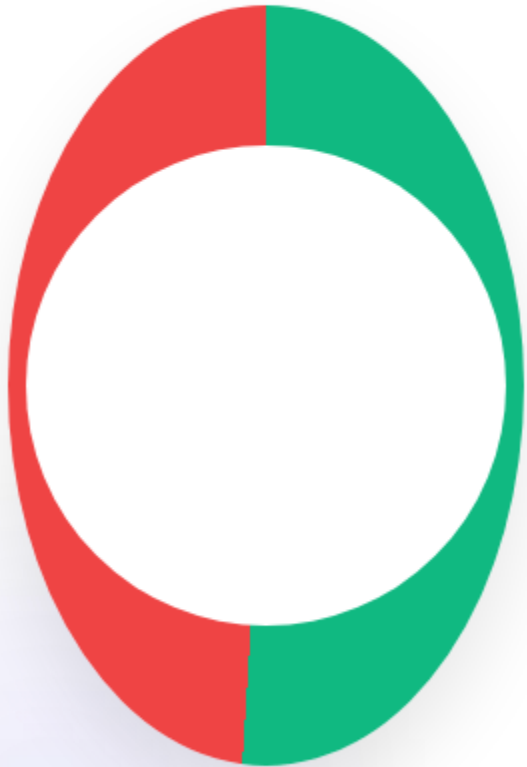
8,005

TOTAL COMMENTS ANALYZED

Massive Scale Efficiency

The Python script processed this entire dataset in less than 5 seconds, identifying critical sentiments that would take a human team days to read.

Sentiment Distribution



■ Positive: 4,062 (51%)

■ Negative: 3,910 (49%)

Critical Insight: Nearly 49% of customers are reporting issues. These 3,910 users require immediate attention to prevent churn.

Project Deliverables



NLP Model

Python script `nlp\_model.py`
for automated daily sentiment
tracking.



Critical List

Isolated list of 3,910 negative
comments for immediate
service recovery.



Master Data

Cleaned CSV dataset ready for
Tableau & PowerBI
Dashboarding.



Questions?

E-commerce Sentiment Analysis Complete

DATA SCIENCE | PROJECT M2